Math 242, S20 **Name:** Answer

Quiz 10

Instructions: **There are totally 3 problems. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (3 points) Compute $\mathcal{L}\{t\cos 3t\}$ using the formula for Differentiation of Transforms $\mathcal{L}\{tf(t)\} = -F'(s)$.

$$\mathcal{L}\{t\cos 2t\} = -\frac{d}{ds}\left[\mathcal{L}\{\cos 3t\}\right]$$

$$= -\frac{d}{ds}\left[\frac{s}{s^2+9}\right]$$

$$= -\frac{(s^2+9)-s(2s)}{(s^2+9)^2}$$

$$= \frac{s^2-9}{(s^2+9)^2}$$

Problem 2. (3 points) Compute $\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+16)}\right\}$ using the convolution property $\mathcal{L}^{-1}\{F(s)G(s)\} = f(t) * g(t)$.

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+16)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} * \mathcal{L}^{-1}\left\{\frac{1}{s^2+16}\right\}$$

$$= 1 * \frac{1}{4}\sin 4t$$

$$= \int_0^t \frac{1}{4}\sin 4\tau \, d\tau$$

$$= -\frac{1}{16}\cos 4\tau \Big|_{\tau=0}^{\tau=t}$$

$$= \frac{1}{16}(1-\cos 4t)$$

Problem 3. (4 points) Compute $\mathcal{L}^{-1}\left\{\tan^{-1}\frac{3}{s+2}\right\}$

$$\mathcal{L}^{-1}\left\{\tan^{-1}\left(\frac{3}{s+2}\right)\right\} = -\frac{1}{t}\mathcal{L}^{-1}\left\{\frac{d}{ds}\left[\tan^{-1}\left(\frac{3}{s+2}\right)\right]\right\}$$

$$= -\frac{1}{t}\mathcal{L}^{-1}\left\{\frac{1}{1+\left(\frac{3}{s+2}\right)^2}\frac{d}{ds}\left(\frac{3}{s+2}\right)\right\}$$

$$= -\frac{1}{t}\mathcal{L}^{-1}\left\{\frac{1}{1+\frac{9}{(s+2)^2}}\cdot\left(-\frac{3}{(s+2)^2}\right)\right\}$$

$$= -\frac{1}{t}\mathcal{L}^{-1}\left\{\frac{-3}{(s+2)^2+9}\right\}$$

$$= \frac{1}{t}\mathcal{L}^{-1}\left\{\frac{3}{(s+2)^2+3^2}\right\}$$

$$= \frac{1}{t}e^{-2t}\mathcal{L}^{-1}\left\{\frac{3}{s^2+3^2}\right\}$$

$$= \frac{1}{t}e^{-2t}\sin 3t$$